PARKEASE - Smart Parking & Slot Booking System

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PARKEASE

City-Wide Parking Marketplace & Management Platform

Find. Reserve. Park.

A unified platform connecting drivers, operators and administrators

Domain: Urban Mobility / Smart Parking / Software Engineering



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Section: F
Subject- LIVE PROJECT
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Urban parking is complex.

PARKEASE makes it simple.

- Growing vehicles and limited parking create search time, congestion and uncertainty
- Drivers need destination-based discovery, comparison and reservation
- Operators need digital inventory, bookings, entry/exit, pricing, staff and revenue tools
- PARKEASE connects Driver, Operator and Admin in one unified platform



PARKEASE IN URBAN MOBILITY

Drivers → **PARKEASE Platform** → **Operators** → **Admin**

A city-wide parking marketplace and management platform **connecting people, spaces and operations.**

Literature Review and Existing System

RESEARCH FOCUS

- Information collection
  Deployment and service dissemination
- Reservations and guidance
  Occupancy and resource management
- IoT work commonly covers occupancy detection, authentication, reservations and online payment

CURRENT PLATFORMS

Park+
provides parking discovery, map view and advance reservation

Arknx
provides driver booking, payment, QR and vendor tooling, positioned around Bengaluru

Parkoza
provides peer-to-peer parking booking and listing across multiple Indian cities



Core parking features exist already; PARKEASE focuses on an integrated product model.

Problem Identification *OR* Limitation of the Existing System

- Driver and operator needs are handled in separate experiences
- Inventory visibility is inconsistent, affecting availability confidence
- operators need integrated workflows for slots, bookings, staff, pricing and revenue
- Destination discovery and operations need a shared, real-time inventory foundation



Project Planning / Proposal

A phased plan to build and deliver **PARKEASE**



FOR DRIVERS

search · compare · select · reserve · pay · QR pass

Find, compare and book parking in just a few simple steps

FOR OPERATORS

facilities · slots · bookings · QR check-in/out · pricing · revenue · analytics

Manage daily operations, optimize pricing and grow revenue

FOR ADMIN

users · operators · parking · bookings · complaints · cities · subscriptions · reports

Oversee the platform, governance and city-wide performance

BUILT FOR CITIES

- PARKEASE Pass for seamless access
- PARKEASE Business for enterprise needs
- Real-time availability you can trust

TECH STACK

- React, Tailwind
- Node, Express, MySQL
- REST APIs, QR and maps integration

IMPLEMENTATION PHASES

01 Foundation → 02 Driver Platform → 03 Operator Platform → 04 Admin Platform → 05 Integration & Backend → 06 Testing & Deployment

PARKEASE
Find. Reserve. Park.

Future Scope

PARKEASE has the potential to scale and evolve into a comprehensive, smart parking ecosystem that transforms urban mobility.



01

Multi-City Expansion

- Scale from the pilot city to multiple cities across India.
- Support city-wise configurations for operators, pricing and policies.



02

Real-Time Parking Integration

- Integrate with IoT sensors and live operator systems for real-time slot availability.
- Provide dynamic pricing based on occupancy and demand.



03

ANPR & Automated Entry/Exit

- Use number plate recognition for seamless and contactless entry and exit.
- Reduce manual effort and improve operational efficiency.



04

Corporate Parking Solutions

- Provide dedicated parking plans for companies, employees and business fleets.
- Enable monthly passes and corporate billing.



05

Event Parking Management

- Support temporary parking management for events, festivals and large venues.
- Enable dynamic capacity, pricing and crowd management.



06

Advanced Analytics & Predictive Insights

- Leverage data for occupancy trends, demand prediction, revenue insights and better city planning.
- Help operators and cities make data-driven decisions.



Reference

- IEEE smart-parking surveys and IoT studies (2017, 2019, 2021, 2023)
- IEEE review of smart parking UI features and real-world examples (2023)
- Park+ official parking page
- Arknx official site
- Parkoza official site

